What’s Wrong With Science?

Session 3: How science is communicated. From science publishing to the popular press.

Professor David Gresham
Fall 2026



Apoorva Mandavilli (New York Times)

WDC TV NEWS

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POPULAR ARTICLES

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Science

Scientists discover first new penguin species in more than 100 years

wdc tv news staff · 16 hours ago · 12 · 6 minutes read

HEALTH

Your daily coffee habit may play a role in dementia risk, study finds

Researchers tracked over 130,000 doctors and nurses for four decades to reach their findings

By Khloe Quill · Fox News

Published February 10, 2026 5:00am EST



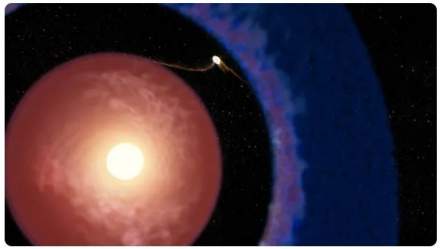
Astronomy > Exoplanets

Add Fox News on Google

Scientists find youngest exoplanet on record — a 'newborn' named Elias

By Chelsea Gohd | Published September 17, 2026

That's one big baby.



Artist's concept depicting Elias 2–24 b, the youngest exoplanet detected to date, still growing within the disk of gas and dust surrounding its young host star.
(Image credit: W. M. Keck Observatory/Adam Makarenko)

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Science News *from research organizations*

Stanford scientists discover the human brain may actually be two separate organs

Date: September 21, 2026

Source: Stanford Medicine

Summary: Scientists have found that the human brain develops from two distinct cellular systems, suggesting that evolution fused together two ancient nervous systems with very different jobs. The discovery also allowed researchers to grow human hindbrain neurons in the lab, opening new possibilities for studying ALS, spinal muscular atrophy, and other brainstem diseases.

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News Article

Scientists Discover Brain Change That May Explain Alzheimer's

PUBLISHED AUG 10, 2026 AT 11:53 AM EDT

UPDATED AUG 11, 2026 AT 11:58 AM EDT



Telephone Rules

1. Read the article quietly (5 minutes)
    a. North Side
    b. South Side

2. Explain the article
    a. North Side to South Side (3 minutes)
    b. South Side to North Side (3 minutes)

3. Explain the finding
    a. South Side (2 minutes)
    b. North Side (2 minutes)

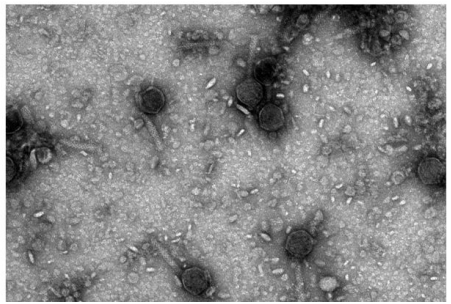
North side read this

The New York Times

The Next Gene-Editing Technology May Also Be the Oldest

Scientists have discovered a powerful gene-targeting system in viruses. Billions of years old, VIPR seems to predate CRISPR and may be even more effective.

Listen · 7:49 min Share full article 41



PDFs

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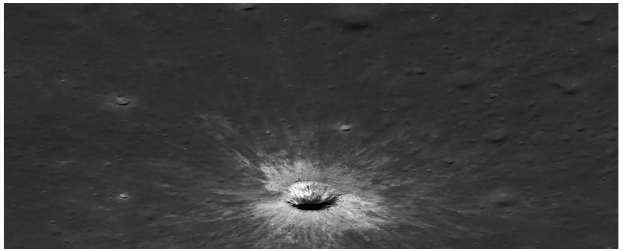
SPACE & COSMOS

The New York Times

What a Massive New Crater Means for Future Moon Bases

At a whopping 728 feet wide, the newfound basin hints at the risks that impact events may pose to planned human outposts.

Listen · 5:31 min Share full article



South Side: What is the article about according to your colleagues?

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North Side: What is the moon base article about?

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Behind the news story is a scientific article

The New York Times

The Next Gene-Editing Technology May Also Be the Oldest

Scientists have discovered a powerful gene-targeting system in viruses. Billions of years old, VIPR seems to predate CRISPR and may be even more effective.

RESEARCH ARTICLES

MICROBIOLOGY

A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas

Peter H. Yoon[1,2]†, Kenneth J. Loi[2,3]†, Zeyuan Terry Zhang[2,3], Trevor A. Docter[1], Santiago C. Lopez[1], Conner J. Langeberg[1,4], Muhammad Moez Ur-Rehman[1,2], Kamakshi Vohra[1,4]‡, Zehan Zhou[1,2], Isabel Esain-Garcia[2,4,5], Marena I. Trinidad[2,6], Honglue Shi[1,4], Ron Boger[2,3], Peter Y. Wang[2,5], Benjamin A. Adler[2], Stephen G. Brohawn[1,4,7], Jennifer A. Doudna[1,2,3,4,5,8,9,10]*

CRISPR-Cas provides RNA-mediated adaptive immunity, but how its first RNA-guided effector arose is unclear. In this study, we report the discovery of Viral Interference Programmable Repeat (VIPR) systems consisting of a Vipr protein ancestral to the earliest CRISPR-Cas effectors and VIPR RNAs (vrRNAs) comprising alternating GGY/NN motifs. Unlike canonical guide RNAs that pair with target nucleic acids through contiguous complementarity, vrRNAs recognize double-stranded DNA through a noncontiguous code in which the variable NN dinucleotides collectively specify a gapped target sequence. Natural vrRNA targets suggest that VIPR systems act against competing phages, and we demonstrate programmable phage defense by redirecting the complex for transcriptional repression. These results suggest that adaptive immunity originated from ancient warfare between viruses, revealing a previously unidentified logic for encoding information in sequence.

The New York Times

What a Massive New Crater Means for Future Moon Bases

At a whopping 728 feet wide, the newfound basin hints at the risks that impact events may pose to planned human outposts.

Science Advances

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RESEARCH ARTICLE | PLANETARY SCIENCE

A new 222-m diameter lunar crater

MARK S. ROBINSON, AARON K. BOYD, PRASUN MAHANTI, MADELEINE R. MANHEIM, EMERSON J. SPEYERER, JULIE D. STOPAR, ROBERT V. WAGNER

Authors Info & Affiliations

SCIENCE ADVANCES • 16 Sep 2026 • Vol 12, Issue 38 • DOI: 10.1126/sciadv.aeh7812

6,629 1

Abstract

Observing the aftermath of recent impact events on the Moon tests models of impact frequency, surface change rates, and cratering physics. The Lunar Reconnaissance Orbiter Camera (LROC) Wide Angle Camera (WAC; 100-meter scale) images the whole Moon each month and has done so since 2010. Comparing images from two months under similar lighting conditions enables automated detection of surface changes. In fall 2025, a routine temporal comparison of WAC image sets revealed a new 222-meter-diameter crater that formed in spring 2024. This is the largest contemporary impact crater found in the Solar System and, according to current models, a once-in-132-year event. Later, a series of LROC Narrow Angle Camera (NAC) images provided high-resolution (meter-scale) photometric and topographic information. These images show that the crater exposed complex geology, inform current models of cratering mechanics, and document widespread surface changes extending more than 1000 crater radii.

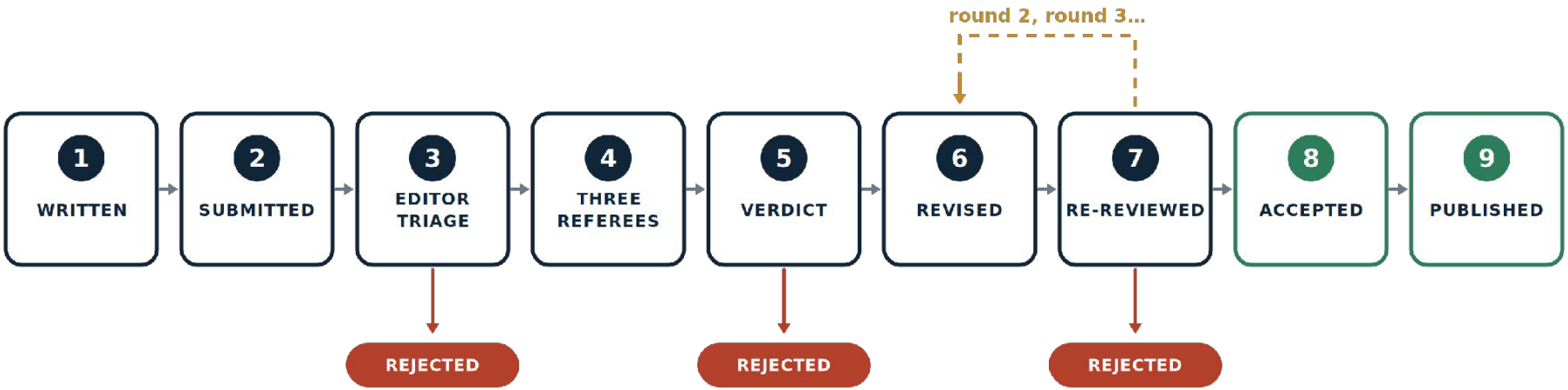
THE PROCESS

Science communication is like playing telephone



How a scientific paper gets published

The whole process



One manuscript. Nine stages. Three places it can die.

THE PROCESS

1 It gets written

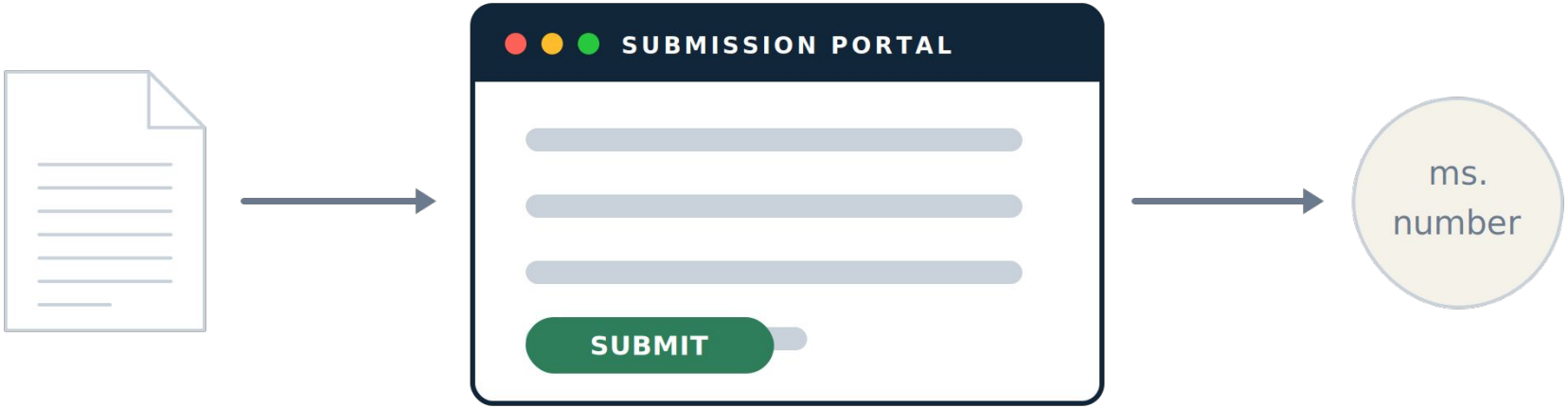
Months to years of work, then weeks of writing.



THE PROCESS

2 It gets submitted

To one journal at a time. Never two.



WHAT IT ACTUALLY LOOKS LIKE

2 It gets submitted

From	mBio editorial office
Date	16 April 2026
Subject	mBio00958-26: Manuscript Received

STEP 2 · SUBMITTED

On April 16, 2026, we received your manuscript “Quiescence improves Candida albicans survival of fungicidal drug exposure” by Ozan Imir, Marina Druseikis, Ying Xie, Yujia Ji, Liam Holt, Judith Berman, and David Gresham.

The manuscript has been assigned the control number mBio00958-26.

THE PROCESS

3 An editor decides

Before any expert reads it.



WHAT IT ACTUALLY LOOKS LIKE

3 An editor decides

From	PLOS Pathogens
Date	26 March 2026 — 27 days later
Subject	Decision: PPATHOGENS-D-26-00515

STEP 3 · DESK REJECTED

Thank you for submitting your manuscript to PLOS Pathogens. As with all papers, your manuscript was reviewed by members of the editorial board.

Based on our assessment, we have decided that the work does not meet our criteria for publication and will therefore be rejected.

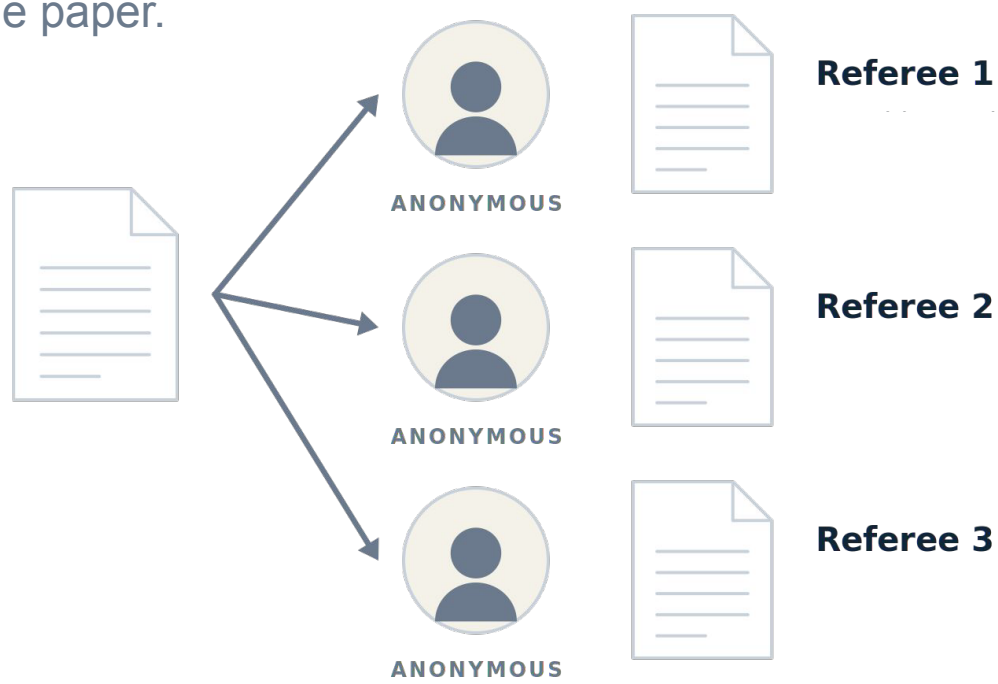
> *... we felt that this manuscript did not provide sufficiently novel findings to warrant external peer review for PLOS Pathogens with the current results.*

Rejected in 27 days. No scientist outside the journal ever read it.

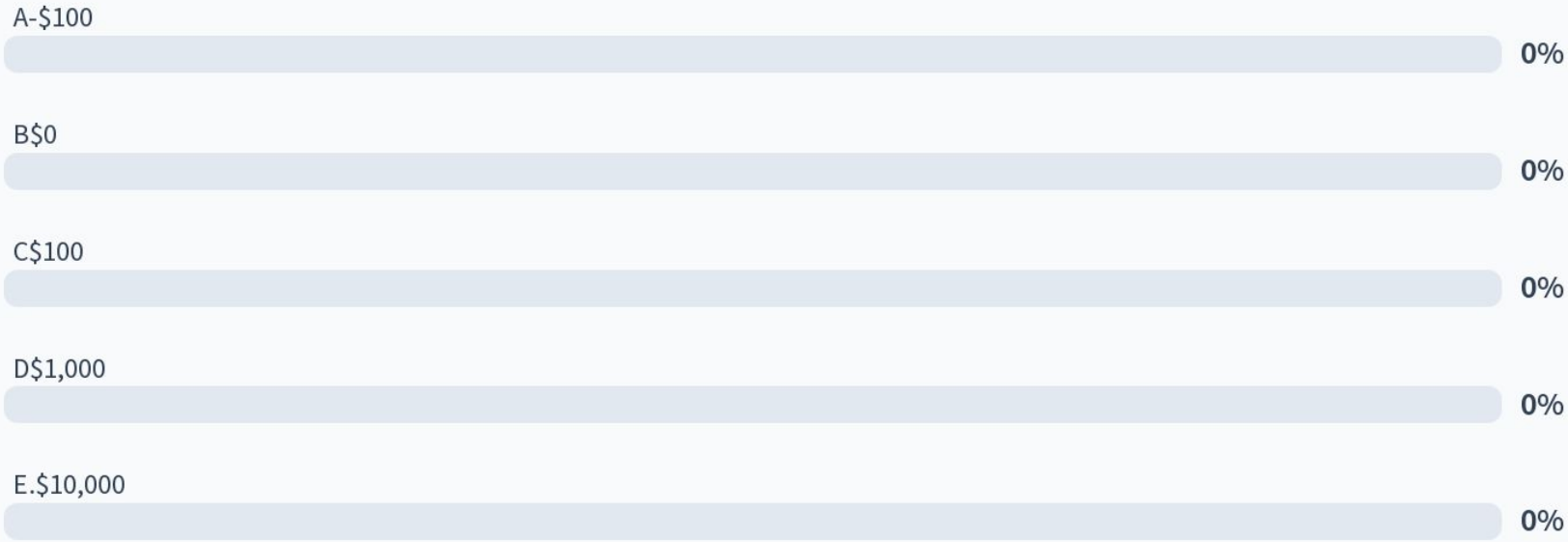
THE PROCESS

4 Three referees read it

Anonymous. Working scientists, like me, who spend ~4-10 hours reading and assessing the paper.



How much does an anonymous peer reviewer typically get paid?



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WHAT IT ACTUALLY LOOKS LIKE

4 Three referees read it

Source mBio referee report
Date 26 May 2026
Author Reviewer #2 (anonymous)

STEP 4 · WHAT A REFEREE SAYS

Overall, this study employs a diverse array of biophysical techniques, RNA sequencing, and single-cell tracking … The data are extensive, and the technical approaches are robust.

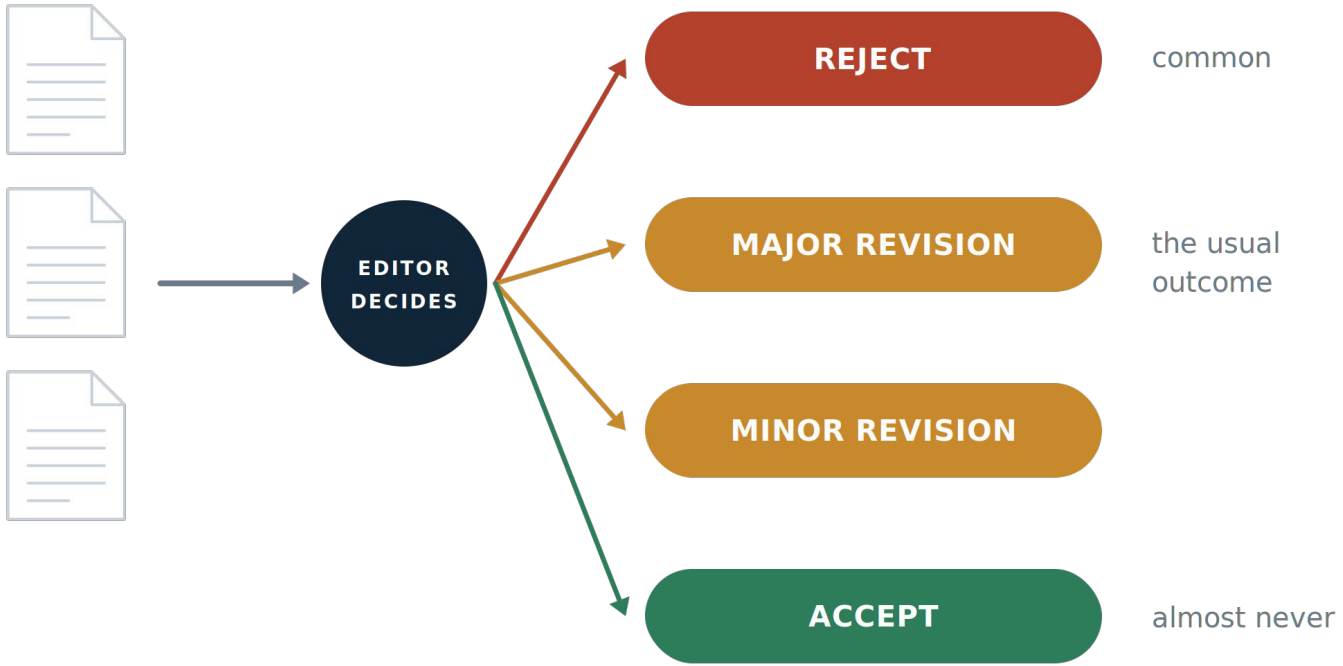
However, the core phenotypic findings … have been extensively documented in C. albicans and other fungal pathogens, thereby limiting the novelty of this work. Furthermore, the manuscript remains largely descriptive and lacks functional validation of the observed phenotypes.

4) Biological replicates are missing from the drug-killing assays shown in Figures 5A-C.

THE PROCESS

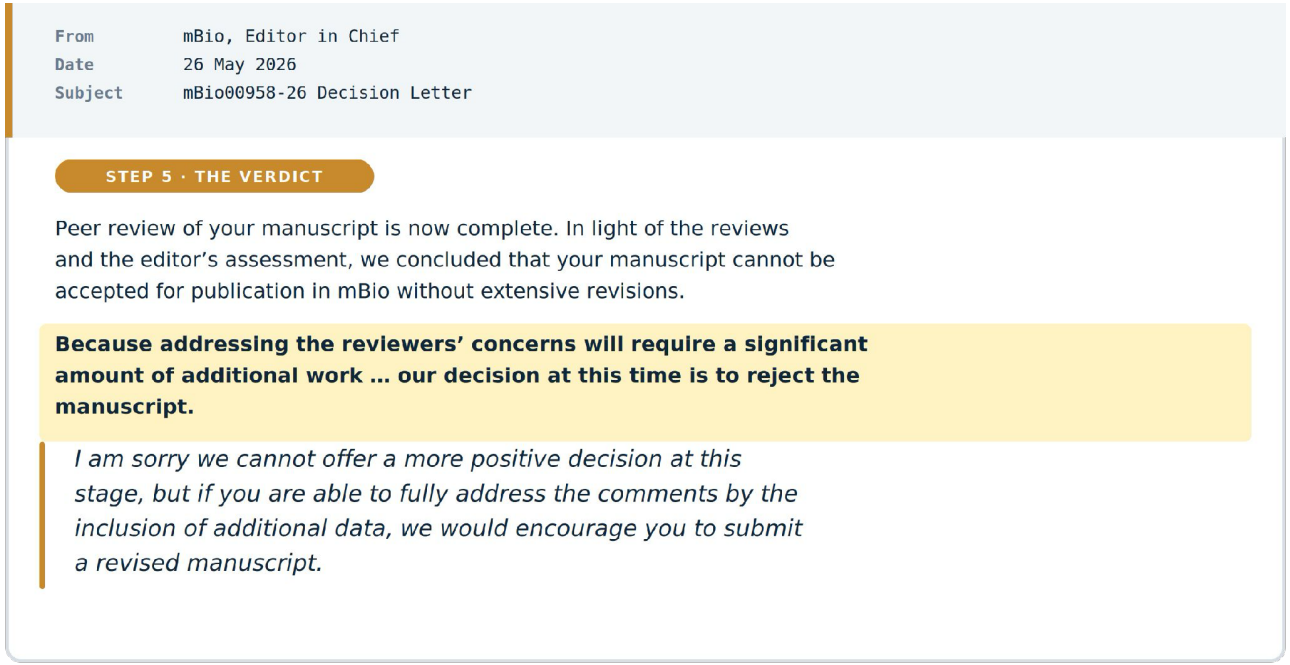
5 The verdict

Outright acceptance is almost unheard of.



WHAT IT ACTUALLY LOOKS LIKE

5 The verdict



From mBio, Editor in Chief
Date 26 May 2026
Subject mBio00958-26 Decision Letter

STEP 5 · THE VERDICT

Peer review of your manuscript is now complete. In light of the reviews and the editor's assessment, we concluded that your manuscript cannot be accepted for publication in mBio without extensive revisions.

Because addressing the reviewers' concerns will require a significant amount of additional work … our decision at this time is to reject the manuscript.

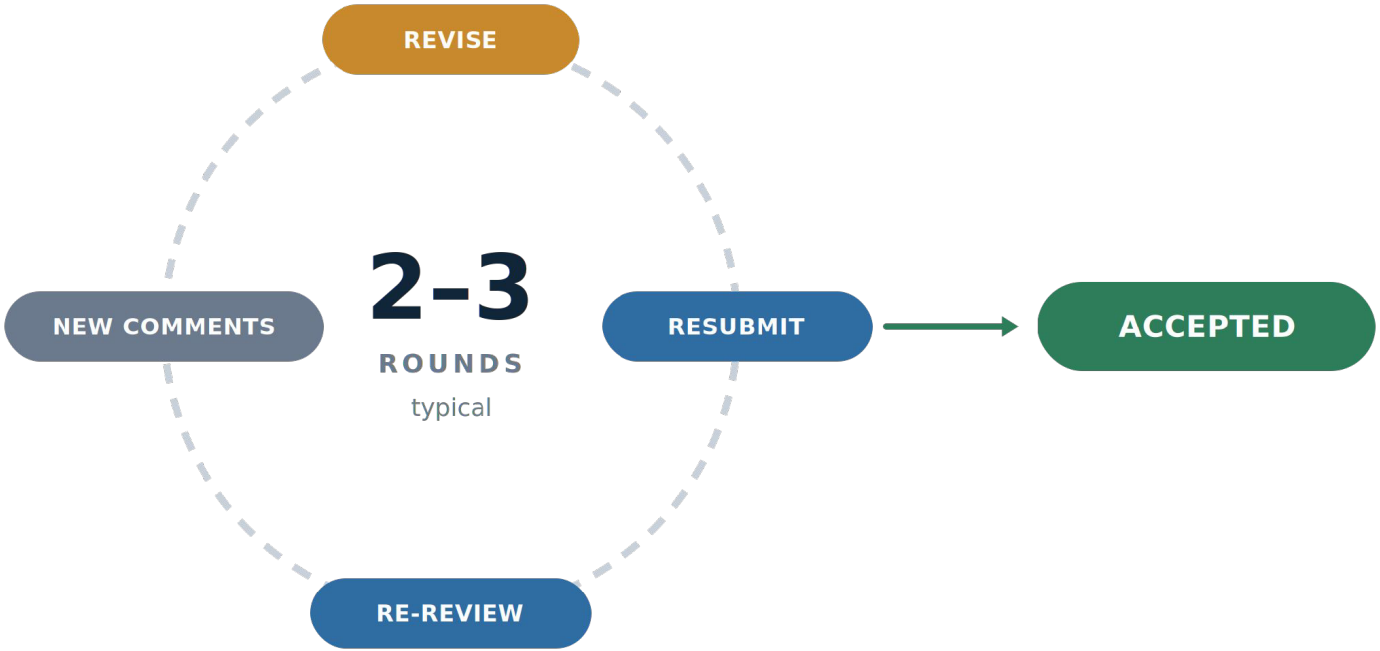
I am sorry we cannot offer a more positive decision at this stage, but if you are able to fully address the comments by the inclusion of additional data, we would encourage you to submit a revised manuscript.

"Reject — but please try again." Both things are true.

THE PROCESS

6 Revise. Resubmit.

New experiments. A letter answering every point.



WHAT IT ACTUALLY LOOKS LIKE

7 It gets re-reviewed

From PLOS Genetics
Date 18 September 2026
Subject Decision: PGENETICS-D-25-01150R2

STEP 7 · ROUND THREE

Reviewer #1: The authors have satisfactorily addressed my remaining concerns. A couple of minor editorial issues remain:

1. PP2ACdc5 should be PP2ACdc55, i.e. a couple of times one “5” is missing.
2. Rim15Δ0 should be rim15Δ0 (several occurrences with a capital “R”).
5. Figure 4 legend: TPS2 should be Tps2.

Round three of a paper first submitted in 2025.

WHAT IT ACTUALLY LOOKS LIKE

8 It gets accepted

From	PLOS Genetics, Academic Editor
Date	18 September 2026
Manuscript	R2 — submitted 2025

STEP 8 · ACCEPTED

In response to the reviewers' requests, the revised manuscript includes additional FACS analysis confirming G1 arrest, as well as cell size analysis. These additional results adequately address the major concerns raised during the previous rounds of review.

Please address the remaining minor points raised by Reviewer 1 … and submit the final revised version for publication.

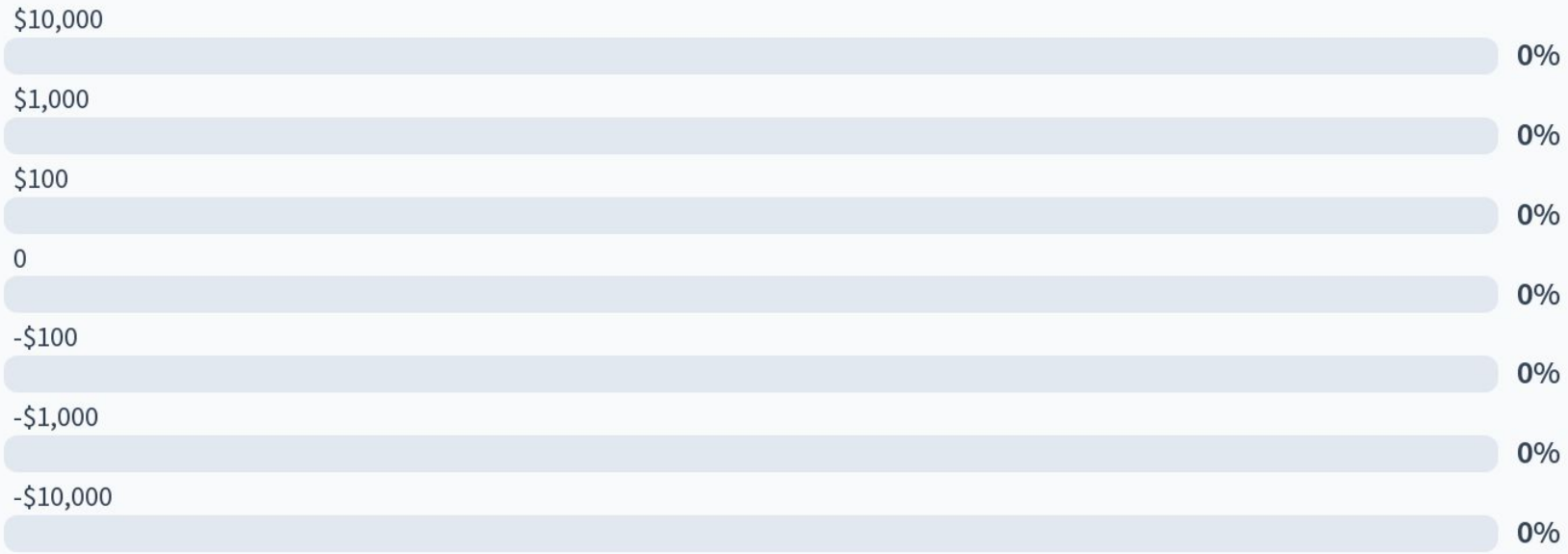
THE PROCESS

9 It gets published

And becomes something anyone can cite.



How much does a scientist get paid to publish their article?



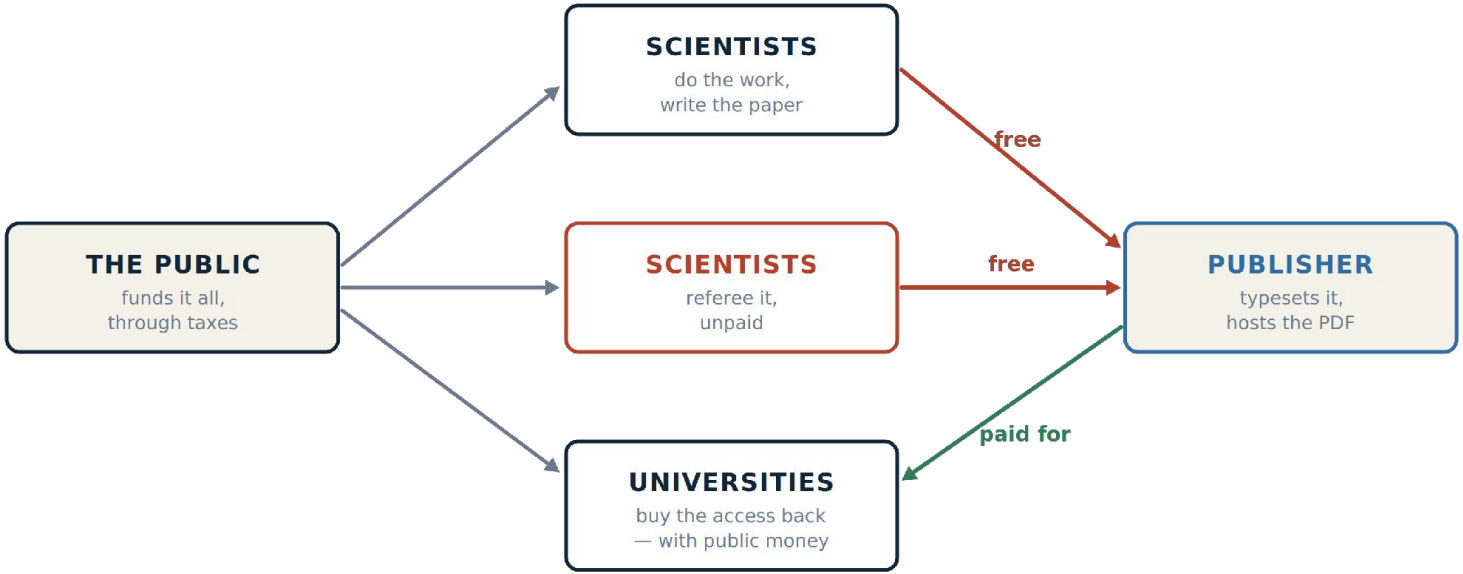
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Everything you read about science came out the other end of this.

Slowly. Anonymously. And only if three strangers said yes.

THE BUSINESS

Who pays, and who works for free



The public funds the research, the salaries, and the subscription. The refereeing is donated.

THE BUSINESS

Who makes money from this

Two companies publish a large share of the world's research.

ELSEVIER

the STM division of RELX plc

£2.71bn

REVENUE

£1.04bn

OPERATING PROFIT

38.1%

MARGIN

Full year 2025

SPRINGER NATURE

Springer Nature AG & Co. KGaA

€1.93bn

REVENUE

€544m

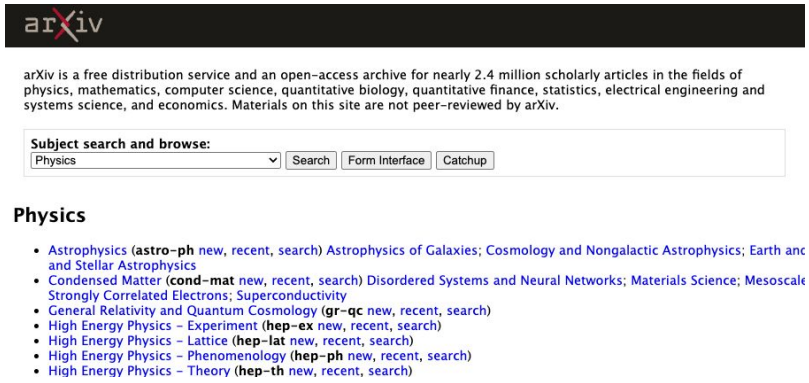
OPERATING PROFIT

28.2%

MARGIN

Full year 2025

Preprints: making papers available without peer review





Preprints: making papers available without peer review

arXiv

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- High Energy Physics - Theory (**hep-th** new, recent, search)

] 21 Sep 2026

Basis function expansions of the two-point correlation function and reconstruction of the cosmological distance scale

Fatima Abd Alrahman[1,2,3], Farnik Nikakhtar[4], Marcello Musso[1,5], Aseem Paranjape[6], and Ravi K. Sheth[7]
[1] *ICTP East Africa Institute for Fundamental Research, University of Rwanda, Kigali – Rwanda*
[2] *Department of Physics, University of Houston, 3507 Cullen Blvd Rm 617, Houston, TX 77204*
[3] *Department of Physics, University of Khartoum, Al-Gama'a Avenue, Khartoum, Sudan*
[4] *Department of Physics, Yale University, New Haven, CT 06511, USA*
[5] *Departamento de Física Fundamental & IUFFyM, Universidad de Salamanca, 37008 Salamanca, Spain*
[6] *Inter-University Centre for Astronomy & Astrophysics, Ganeshkhind, Post Bag 4, Pune 411007 – India and*
[7] *Center for Particle Cosmology, University of Pennsylvania, 209 S. 33rd St., Philadelphia, PA 19104 – USA*
(Dated: September 23, 2026)

The shape of the correlation function of galaxy pairs can be used as a cosmological distance indicator. Most estimators of this shape require that one bin the pair counts, and considerable effort is expended in finding the optimal bin size. Cosmological constraints are then derived, in a separate step, by fitting to the binned pair counts. By projecting the pair counts on a suitable basis set, it is possible to estimate the pair correlation function without binning. Recent work suggests that, in the context of estimating the cosmological distance scale from galaxy clustering data, polynomials and half-integer generalized Laguerre functions are particularly useful bases. We show how to use these bases to measure the correlation function and estimate the reconstructed distance scale in a single step. Our analysis includes the effects of shot-noise and cosmic variance, and highlights the benefits of projecting pair counts onto orthogonal basis functions.

Press releases: Universities pitch their stories

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RESEARCH, PATIENT CARE, PRESS RELEASES

Rutika Mehta, MD, MPH, Joins NYU Langone's Perlmutter Cancer Center to Expand Upper GI Cancer Care and Research

She will grow personalized care and clinical trials for upper GI cancers.



September 21, 2026

PATIENT CARE, LOCATIONS, PRESS RELEASES

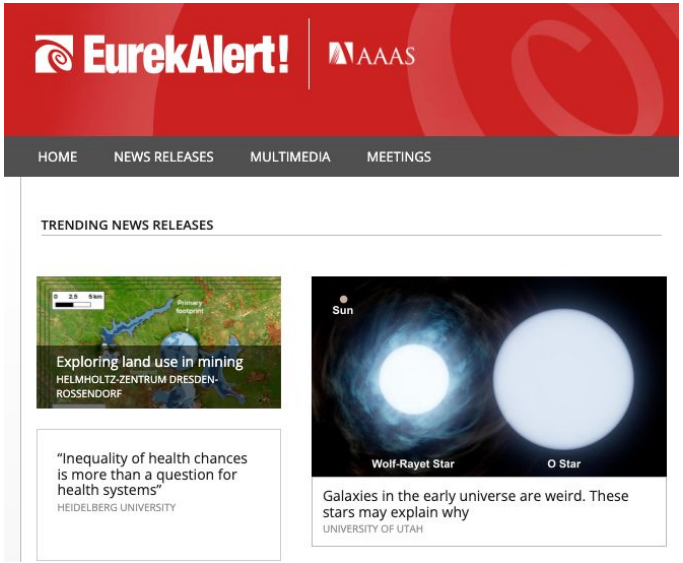
Family Health Centers at NYU Langone's Annual Back-to-School Drive Helps Prepare Students Across Brooklyn and Long Island

More than 1,000 backpacks and school supplies were distributed.



September 18, 2026

Press releases: aggregation services



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